B. Tech. Degree III Semester Examination November 2013

ME 1304 FLUID MECHANICS

(2012 Scheme)

Time : 3 Hours | Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 x 5 = 40)

I. (a) Distinguish between dynamic viscosity and kinematic viscosity.

(b) Explain the terms metacentre and metacentric height.

(c) Explain minor losses in pipes.

(d) Briefly explain significance of Moody's chart.

(e) A fluid flow is given by $V = 8x^3i-10x^2yj$. Verify the flow is rotational or irrotational.

(f) What is magnus effect?

(g) Define displacement thickness, momentum thickness and the energy thickness.

(h) Briefly explain skin friction drag.

PART B

(4 x 15 = 60)

II. (a) Briefly explain various properties of fluids. (8)

(b) A flat plate of area $1.5x10^6$ mm^2 is pulled with a speed of 0.4m/s relative to another plate located at a distance of 0.15mm from it. Find the force and power required to maintain this speed, if the fluid separating them is having viscosity as 1 poise. (7)

OR

III. (a) Briefly describe (i) Stream lines (ii) Stream tubes (iii) Streak lines (iv) Path lines (8)

(b) What is Reynold transport theorem and fluid dynamics concept of control volume. (7)

IV. (a) Derive Hagen Poiseuille Formula. (8)

(b) An oil of viscosity $0.1NS/m^2$ and relative density 0.9 is flowing through a circular pipe of diameter 50mm and of length 300m. The rate of flow of fluid through the pipe is 3.5 litre/s. Find the pressure drop in a length of 300m. (7)

OR

V. (a) Briefly describe an orifice meter and a pitot tube. (8)

(b) A pitot-static tube placed in the centre of a 300mm pipe line has one orifice pointing upstream and other perpendicular to it. The mean velocity in the pipe is 0.80 of the central velocity. Find the discharge through the pipe if the pressure difference between the two orifices is 60mm of water. Take the co-efficient pitot tube as $C_V = 0.98$. (7)

VI. In a two dimensional incompressible flow, the fluid velocity components are given by $u = x - 4y$ and $v = -y - 4x$. Show that the velocity potential exists and determine its form. Find also the stream function. (15)

OR

VII. (a) Briefly describe doublet. (5)

(b) A point P(0.5,1) is situated in the flow field of a doublet of $5m^2/s$. Calculate the velocity at this point. (10)

VIII. (a) Briefly explain the boundary layer procedure (5)

(b) Find the displacement thickness, the momentum thickness and energy thickness for the velocity distribution in the boundary layer given by $\frac{u}{U} = \frac{y}{\delta}$, where u is the velocity at a distance y from the plate and $u = U$ at $y = \delta$, where $\delta =$ boundary layer thickness. Also calculate the value δ^* / θ. (10)

OR

IX. (a) Explain V*on* Karman momentum integral equation. (6)

(b) Explain laminar boundary layer and turbulent boundary layer with sketch. (9)

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